

Interested amateurs. The true target audience of this book is interested amateurs. Algorithms touch nearly every part of life, so everyone should be able to find at least something in this book that enhances their appreciation of the world around them.

About This Book

This book does not cover every aspect of every extant algorithm; it's meant only as an introduction. After reading it, you will have a solid grasp of what an algorithm is, know how to write code to implement important algorithms, and understand how to judge and optimize algorithms' performance. You will also be familiar with many of the most popular algorithms professionals use today. The chapters are organized as follows:

Chapter 1: Problem-Solving with Algorithms, in which we tackle the problem of how to catch a ball, find evidence for a subconscious algorithm governing human behavior, and discuss what that teaches us about the utility of algorithms and how to design them.

Chapter 2: Algorithms in History, in which we travel around the world and through history to find out how ancient Egyptians and Russian peasants multiplied numbers, how the ancient Greeks found greatest common divisors, and how medieval Japanese scholars created magic squares.

Chapter 3: Maximizing and Minimizing, in which we introduce gradient ascent and gradient descent. These simple methods for finding the maxima and minima of functions are used for optimization, an important goal of many algorithms.

Chapter 4: Sorting and Searching, in which we present fundamental algorithms for sorting lists and searching for elements within them. We also introduce how to measure the efficiency and speed of algorithms.

Chapter 5: Pure Math, in which we concern ourselves with purely mathematical algorithms, including those for generating continued fractions, calculating square roots, and generating pseudorandom numbers.